Supplementary Information for "FALI: Mechanistic Decomposition of Conditional Attention-Head Causal Usefulness"

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Abstract

This Supplementary Information provides additional technical detail for the main manuscript, including Owen attribution details, validation chronology, full GPT-2 tables, robustness and gauge-sensitivity results, and the Pythia induction protocol.

Keywords: FALI, supplementary information, mechanistic interpretability

1 Standard Owen attribution and intervention identities

FALI uses five binary factors $N = \{T, R, O, V, C\}$ and the prospectively fixed coalition partition

$$\Pi = \{\text{SCORE} = \{T, R, O\}, \text{VALUE} = \{V\}, \text{CONTEXT} = \{C\}\}.$$

Each coalition $K \subseteq N$ indexes the hybrid state in which members of K take their perturbed-state values and all remaining factors take their intact-state values. The characteristic function is $v(K) = U(\mathbf{1}_K)$, where utility is the candidate-present task metric minus the candidate-zero task metric in the same surrounding context.

The standard Owen value averages marginal contributions over all permutations that keep members of each block contiguous. For the partition above there are $3! \times 3! = 36$ admissible permutations. Efficiency gives

$$\phi_T + \phi_R + \phi_O + \phi_V + \phi_C = v(N) - v(\varnothing) = \Delta_h.$$

The quotient game on SCORE/VALUE/CONTEXT satisfies

$$\Phi_{\mathrm{SCORE}} = \phi_T + \phi_R + \phi_O.$$

These identities allocate interaction under the specified counterfactual game. They do not identify unique natural pathways.

2 Method-validation chronology and controlled ground truth

The attribution implementation was validated before substantive pretrained model experiments. An initial fixed block-order implementation was rejected because it was not the standard Owen value; EF-0R restored all 36 admissible permutations and ordinary Shapley attribution on the quotient game. A subsequent implementation review identified a batch-shared target partition and insufficiently independent endpoint checks in EF-1; EF-1R introduced per-example batched partitions and independent direct endpoint validation before pretrained model experiments began. The superseded implementations are retained only for audit history.

2.1 Single-factor synthetic systems

Table 1 Exact single-factor validation values. In each row only the manipulated factor received attribution up to floating-point error.

Manipulated factor	ΔU	Maximum unintended attribution
T	0.2689520684	0
R	-0.4940997625	0
O	0.2134890065	0
V	0.1925102705	0
C	0.7112345942	0

The largest intended-channel discrepancy was 1.11×10^{-16}. A pure $T \times C$ construction had $\Delta U = 1.999999999996$ with $\phi_T = 0.999999999998$, $\phi_C = 0.999999999998$, and $\phi_R = \phi_O = \phi_V = 0$.

2.2 Random-transformer implementation invariants

The random TransformerLens validation model used two layers, two heads per layer, $d_{\mathrm{model}} = 32$, $d_{\mathrm{head}} = 16$, $d_{\mathrm{mlp}} = 64$, context length 8, vocabulary size 48, FP32 CPU execution, and seed 4242. The primary and candidate were L0H0 and L1H0.

Independent ordinary forwards gave

$$E_A = 0.4550457001,$$
$$E_B = -0.0856835023,$$

$$\Delta_h = -0.5407292023.$$

The all-intact and all-perturbed FALI endpoint cells reproduced these quantities with zero recorded discrepancy. Score-to-pattern reconstruction, pattern and value reconstruction of z, masked probability, candidate-output injection, non-candidate isolation, live-leak checks (Q, V, and Q+V), and quotient-game reconciliation all had zero recorded discrepancy at the reported precision.

3 GPT-2 IOI confirmation

3.1 Design and prospectively frozen fine-channel hypotheses

The canonical cohort used GPT-2 Small, the primary Name Movers L9H6, L9H9, and L10H0 as the perturbed set, candidates L10H2 and L10H10, and architectural controls L9H7 and L9H0. The design contained 15 low-level templates crossed with BABA/ABBA structure, 30 prompts per stratum, and seeds 42, 1, 8, and 22, for 3600 prompts.

The EF-4 confirmation plan was committed before canonical fine-channel execution and was recorded in the run configurations. The four primary directional hypotheses were

$$\phi_T > 0,\ \phi_R < 0$$

for both L10H2 and L10H10. ϕ_O was descriptive. Point estimates equally weighted 30 strata within each seed and then averaged across the four seeds. Intervals used 10,000 hierarchical bootstrap draws over template clusters. Confirmation also required directional agreement in every seed, broad stratum support, and a sensitivity that excludes template 0.

3.2 Canonical exact estimates

The EF-4 fine-score gate passed for both heads. The fine-score signs were in the predicted direction in all four seed means and all 30 stratum means for both channels. The template-0-excluded analysis retained confidence intervals excluding zero in the predicted directions.

The broad functional effects used to classify recruitment were measured in the corrected historical canonical analysis. L10H2 had conditional effect 0.382239 (95% CI $[0.340647, 0.423865]$), while L10H10 had conditional effect -0.078890 ($[-0.111127, -0.045726]$). Positive functional recruitment therefore holds for L10H2 but not L10H10.

4 Alternative primary-intervention schemes

EF-5 tested matched-mean and matched-resample alternatives. The four frozen fine-channel directions were retained under both alternatives.

All 30 strata retained the frozen direction for each reported channel. These channel-level results do not imply an overall EF-5 pass. The preregistered experiment-level gate failed because the zero-reference damage-drop estimate was -0.318825 with 95% CI

Table 2 Canonical GPT-2 FALI results. Intervals are hierarchical 95% bootstrap confidence intervals.

Head	Quantity	Mean	CI low	CI high
L10H2	conditional gain	0.298451	0.262940	0.335184
	Δ target attention	0.279460	0.261643	0.295510
	ϕ_T	0.782806	0.721856	0.838338
	ϕ_R	-0.341419	-0.371673	-0.310909
	ϕ_O	0.003864	0.002363	0.005329
	ϕ_V	-0.000349	-0.000810	0.000104
	ϕ_C	-0.146451	-0.164685	-0.128324
	Φ_{SCORE}	0.445252	0.408802	0.477107
L10H10	conditional gain	-0.263442	-0.301914	-0.224875
	Δ target attention	0.310250	0.285276	0.334539
	ϕ_T	0.328524	0.300205	0.355043
	ϕ_R	-0.125907	-0.138610	-0.113054
	ϕ_O	0.003634	0.002586	0.004619
	ϕ_V	0.001556	0.001019	0.002090
	ϕ_C	-0.471249	-0.514272	-0.427467
	Φ_{SCORE}	0.206251	0.185089	0.225661

Table 3 EF-5 fine-channel robustness under the tested alternatives.

Scheme	Channel	Mean	CI low	CI high
matched mean	L10H2 ϕ_T	0.440043	0.408205	0.469904
	L10H2 ϕ_R	-0.165506	-0.178350	-0.152149
	L10H10 ϕ_T	0.187863	0.172625	0.202101
	L10H10 ϕ_R	-0.081585	-0.088512	-0.074357
matched resample	L10H2 ϕ_T	0.265058	0.245147	0.285338
	L10H2 ϕ_R	-0.047390	-0.057199	-0.038001
	L10H10 ϕ_T	0.085806	0.077595	0.094128
	L10H10 ϕ_R	-0.014964	-0.018855	-0.011402

$[-0.395660, -0.243713]$, violating the required positive reference criterion. The correct interpretation is therefore ROBUST for the tested fine-channel alternatives and FAIL for the overall EF-5 gate.

5 Pythia-160M induction transfer

5.1 Prospective design

EF-6 was the one authorized cross-setting transfer test. Discovery used seed 42 only and was prohibited from inspecting FALI attributions. The frozen selection rule chose L5H0 as the primary and L8H2 as the candidate. Confirmation used seeds 1, 8, and 22 with offsets 8, 16, and 24 and 128 examples per offset per seed. The task metric was target-token natural-log probability. The transfer hypotheses predicted $\phi_T > 0$ and $\phi_R < 0$.

The repeated-token construction used base length 32, target token x_{j+1}, final query $q = 32+j$, and target key $k = j+1$. The control replaced the first 32-token block with a

disjoint token block. Estimates weighted offsets equally within seed and seeds equally; confidence intervals used 10,000 deterministic bootstrap draws within seed–offset cells.

5.2 Held-out exact results

Table 4 Pythia-160M confirmation results.

Quantity	Mean	CI low	CI high
induction advantage	14.017091	12.280371	15.939234
primary drop	1.145992	0.886752	1.538771
conditional effect	0.621197	0.455912	0.860486
conditional gain	0.485653	0.326486	0.721815
ϕ_T	-0.143737	-0.247435	-0.084085
ϕ_R	0.117800	0.058225	0.219125
ϕ_O	-0.000270	-0.000847	0.000256
ϕ_V	-0.002099	-0.002762	-0.001557
ϕ_C	0.513958	0.354460	0.752149

G0 (induction behavior), G1 (primary causal importance), and G2 (positive conditional recruitment) were confirmed. G3 and G4 were not confirmed because the target and rival signs were opposite to the preregistered transfer hypothesis. Accordingly, EF-6 is FAIL. This is a failed mechanism-transfer hypothesis, not a failed recruitment result and not confirmation of a universal opposite mechanism.

6 Softmax-gauge sensitivity (prespecified row-centering)

Native-logit FALI is the primary analysis. The following tables report the preregistered zero-mean (row-centering) gauge sensitivity (preregistration commit `e60f81acfff8fc9dd4f924ebb1c087ec89734c05`; executed results commit `43135198065e4a13acb241c677a0965e2165b988`). Source report: `results/extended_fali/gauge_sensitivity/GAUGE_SENSITIVITY_REPORT.md`.

Cross-setting fine contrast: PRESERVED. Interpretation code: A (fine-channel sign conclusions robust to the prespecified zero-mean gauge sensitivity only; not universal gauge invariance; not magnitude invariance).

Table 5 GPT-2 EF-4 L10H2 native vs row-mean gauge sensitivity.

Metric	Native mean	Row-mean mean	Paired Δ
ϕ_T	0.782806	0.643597	-0.139209
ϕ_R	-0.341419	-0.202220	0.139199
ϕ_O	0.003864	0.003875	1.02×10^{-5}
ϕ_V	-0.000349	-0.000349	0
ϕ_C	-0.146451	-0.146451	0
Φ_{SCORE}	0.445252	0.445252	≈ 0
conditional gain	0.298451	0.298451	0

Table 6 GPT-2 EF-4 L10H10 native vs row-mean gauge sensitivity.

Metric	Native mean	Row-mean mean	Paired Δ
ϕ_T	0.328524	0.305908	-0.022616
ϕ_R	-0.125907	-0.103273	0.022635
ϕ_O	0.003634	0.003615	-1.91×10^{-5}
ϕ_V	0.001556	0.001556	0
ϕ_C	-0.471249	-0.471249	0
Φ_{SCORE}	0.206251	0.206251	≈ 0
conditional gain	-0.263442	-0.263442	0

Table 7 Pythia EF-6 L8H2 native vs row-mean gauge sensitivity.

Metric	Native mean	Row-mean mean	Paired Δ
ϕ_T	-0.143737	-0.047046	0.096690
ϕ_R	0.117800	0.021088	-0.096712
ϕ_O	-0.000270	-0.000247	2.22×10^{-5}
ϕ_V	-0.002099	-0.002099	0
ϕ_C	0.513958	0.513958	0
Φ_{SCORE}	-0.026206	-0.026206	≈ 0
conditional gain	0.485653	0.485653	0

Fine-channel decisions: L10H2, L10H10, and L8H2 all PRESERVED under the pre-registered sign criteria. Magnitudes change (especially Pythia). No claim of complete gauge invariance is made.

7 Reproducibility and audit trail

The experiment repository contains the prospective analysis plans, run configurations with preregistration commit identifiers, exact balanced means, bootstrap intervals, gate files, validation reports, and machine-readable raw and aggregate outputs. The central evidence paths are:

- `results/extended_fali/ef1_synthetic/validation_report.json`
- `results/extended_fali/ef4_canonical/aggregate/`
- `results/extended_fali/ef5_ablation_robustness/aggregate/`
- `results/extended_fali/ef6_pythia_induction/aggregate/`
- `results/extended_fali/gauge_sensitivity/`
- `results/revision_phase4/full/aggregate/`

No additional model, task, threshold, or replacement pair was introduced after the reported EF-5 or EF-6 failures. Hybrid-state results should be interpreted as outcomes of the stated counterfactual intervention game; the surrounding context factor remains coarse and can aggregate multiple external computations.